GABLS

The GEWEX Atmospheric Boundary-Layer Study (GABLS)

The GEWEX Atmospheric Boundary-Layer Study (GABLS) focus on the representation of stable boundary layers in atmospheric models (Holtslag, 2006). One of the main goals of GABLS is to provide a mondial platform for the atmospheric boundary layer research community through the organisation of model intercomparisons. Here we focus on single column models (SCM's), which can be both research models and SCM's derived from operational weather and climate models. Two SCM intercomparison case studies have been performed so far. One highly idealised case over snow with prescribed surface temperature (Cuxart et al., 2006) and a second case based on observations taken during the CASES 99 stable boundary layer experiment also with prescribed surface temperature (Svensson and Holtslag, 2007).

In these first two studies it was found that especially the complexity of real world boundary conditions and the lack of interaction with the surface makes it difficult to confront the models with observed evaluation parameters. A reasonable ideal 3rd GABLS case was found in the long observational dataset of the meteorological site Cabauw in the Netherlands . To make comparison with observations possible care was taken to prescribe realistic advective tendency terms to the SCM's. These were estimated from both local observations and hind casts of several 3D NWP models. The specific characteristics of the Cabauw site with its flat topography (van Ulden and Wieringa, 1995; Beljaars and Bosveld, 1997) makes it well suited to study decoupling around sunset, inertial oscillation and low level jet and the morning time transition to convective conditions (Angevine et al. 2002).

Literature

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- Beljaars A. C. M. and F. C. Bosveld (1997). Cabauw data for the validation of land surface parameterization schemes. J. of Climate, 10, 1172-1193.
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- Cuxart J., A.A.M. Holtslag et al. (2006). Single-column model intercomparison for a stably stratified atmospheric boundary layer. Bound.-Layer Meteor., 118, 273-303.
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Calendar

- 26-27 June 2009 GABLS Workshop, Boulder, Colorado, USA
- 15 Nov 2008 Dead line for sending results.
- 29 Sep 2008 Presentation of intermediate results at the 8th Annual Meeting of the

- 29 Sep 2008 Presentation of intermediate results at the 8th Annual Meeting of the EMS/7th ECAC, 29 Sep - 3 Oct 2008, Amsterdam, The Netherlands.
- 01 May 2008 Final date for sending results to be incorporated in preliminary analysis to be presented at the AMS 18th Symposium on Boundary Layer and Turbulence, 9-13 June 2008 Stockholm, Sweden.
- 01 Feb 2008 Release of case set up

Updates of the Site

- 23-Sep-2008 Refinement of model output definition. Use time dimension as unlimited or at least as the slow varying dimension.
- 09-Sep-2008 Refinement of model output definition. "Soil heat conductance" changed to "Soil heat flux".
- 19-May-2008 Refinement of model output definition. _FillValue atrribute; sign convention surface parameters; Physical parameters in NetCdf type float.
- 24-Apr-2008 Added the term "Variables" between the dimension section and variable section in the model output description.
- 21-Feb-2008 Included more soil and vegetation information in the case set-up. And asked operational models to run in operational resolution.
- 25-Jan-2008 Creation

More on this topic

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- GABLS1 SCM
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